TFM Multimodal Integration Spatial Transcriptomics



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Work Area: Omics Data Analysis

Master's degree in Bioinformatics and Biostatistics

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Final Work Card

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Abstract

This Master's Thesis (TFM) focuses on the field of Omics Data Analysis, specifically addressing the integration of spatial transcriptomics and single-cell RNA sequencing (scRNA-seq). The ultimate goal is to develop and implement a Deep Learning architecture combining Graph Neural Networks (GNN) and Multimodal Variational Autoencoders (VAE). This architecture will be explicitly conditioned by biological sex to model cellular heterogeneity in cardiovascular tissues. Upon completion, the project will deliver a trained model, an extraction of interpretable biomarkers using eXplainable Artificial Intelligence (XAI), and an interactive dashboard for visualization.

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Chapter 1

PEC1 - Introduction

This Master's Thesis proposes developing a Geometric Deep Learning architecture conditioned on biological sex for the analysis of spatial transcriptomics (ST) in human atherosclerotic plaques [4, 5]. By combining a **Graph Attention Network (GAT)** with a **Conditional Multimodal Variational Autoencoder (C-mVAE)**, the model will identify spatially defined cellular niches exhibiting sex-specific transcriptional programs, extracting biomarkers via XAI techniques (GNNExplainer + SHAP) [12]. The deliverables include a trained model, a prioritised list of sex-specific biomarkers, and an interactive visualisation application [8].

1.1 Context and justification of work

Precision medicine relies on capturing not only which genes are expressed, but also the precise spatial and cellular contexts in which their activation occurs. In cardiovascular disease (CVD), the leading cause of mortality among women worldwide [2], the specific molecular mechanisms governing female vascular tissue remain largely unknown due to historical underrepresentation in clinical trials and functional genomics.

Standard bioinformatics workflows frequently ignore biological sex or treat it merely as a confounding factor to be removed during batch correction, thereby obscuring sex-specific pathophysiological programs and reducing the efficacy of targeted therapies in women. Addressing this fundamental limitation, a recent review in *Atherosclerosis* (2023) emphasised that **"future research would require higher inclusion rates of women and well-defined sex-stratified multi-omics analyses"** [1].

This project directly responds to this critical gap by introducing an algorithmic solution that incorporates biological sex not as noise, but as an explicit, structural conditioning variable within the neural network architecture which representing a key methodological advancement for the discovery of spatially resolved, sex-specific cardiovascular biomarkers.

1.2 Motivation

The primary motivation stems from the desire to bridge over a decade of professional experience in Machine Learning and MLOps engineering with frontier challenges in bioinformatics and health data science. Academically, this Master's Thesis constitutes a fundamental pillar and an

essential proof of concept in preparation for future Ph.D. studies in computational biology or bioinformatics, with a focus on biopharmaceutical applications.

1.3 Work objectives

- Design and train a Deep Learning architecture (GAT + C-mVAE) for spatial transcriptomics data.
- Extract sex-specific ligand-receptor communication biomarkers using XAI techniques.
- Deploy the results in an interactive web application that allows the exploration of cellular niches.
- In scope: Use of public datasets (GEO), computational deconvolution, and predictive modeling.
- Out of scope: Collection of in vivo or wet lab clinical samples, experimental clinical validation.

1.3.1 Main Objective

The explicit inclusion of biological sex as a conditioning variable in geometric and generative architectures significantly improves the identification and explainability of spatial cellular niches and transcriptomic biomarkers.

1.3.2 Research Questions

1. How does the deconvolution of transcriptomic spots using a sex-disaggregated scRNA-seq atlas affect the quality of the initial spatial signal?
2. Is it possible to stabilize the joint training of GNNs and VAEs by introducing Feature-wise Linear Modulation (FiLM) layers conditioned by sex [10]?
3. Which cellular communication axes (ligand-receptor) are prioritized by XAI algorithms [12] in female versus male samples?

1.4 Focus and method followed

Research strategy: A quantitative and experimental research strategy will be adopted, evaluating algorithmic performance against baseline models (such as STAGATE [8]) using established metrics (ARI, NMI).

Development methodology: An iterative approach based on prototyping will be used, implementing reproducible Machine Learning operations (MLOps) pipelines [9].

Tools: PyTorch Geometric (network architecture), Squidpy and Scanpy [6] (spatial and scRNA-seq processing), GitHub (version control), Docker (isolated environments), and Streamlit or R Shiny (final dashboard creation).

1.5 Impact on sustainability, ethics-social and diversity

The scientific core of this project is, by definition, the integration of the gender perspective and biological sex into precision medicine, seeking to correct the biases of generic models. Additionally:

1. SAGER (Sex and Gender Equity in Research) guidelines [16] will be applied for the correct reporting of variables in the analyses.
2. All technical documentation and the final report will use inclusive and non-sexist language.
3. The visualization dashboard will be designed and tested to ensure that differences in results across various biological profiles are presented clearly and without presentation biases.

1.6 Methodological Design and Consistency

1.6.1 Pipeline Overview

Phase	Name	Selected Method	Considered Alternative	Selection Criteria
1	Ingestion & QC	SpaceRanger + Squidpy	STARsolo, Cellranger	GEO compatibility, documentation
2	Deconvolution	SPOTlight (scRNA GSE210572)	RCTD, CARD	Sex-stratified ref. available [15]
3	Baseline GNN	STAGATE (GAT)	SpaGCN [7], BayesSpace	Top ARI performance in benchmarks [8, 7]
4	Sex-Strat. GNN	PyTorch Geometric + FiLM	Original Proposal	Literature gap verified
5	Integration	MOFA+ → C-mVAE PoE	DIABLO, scVI	MOFA+ validated in CVD [13]
6	XAI & Viz.	GNNExplainer + SHAP + Streamlit	LIME, IntegratedGrad [12]	Native graph support

1.6.2 Datasets Used

Dataset	Source	Access	Role in the Model
GEO GSE233025	Campos et al. (2025) [5]	Open, Immediate	Expression matrix ($\mathbf{X}$) + Spatial coordinates ($\mathbf{A}$)
HCA Heart (Visium)	Human Cell Atlas [4]	Open (CZI)	Cross-validation
GEO GSE210572	Sukhavasi et al. (2025) [3]	Open, Immediate	Sex-stratified scRNA-seq reference
GTEx v10	GTEx Portal	Open	Epigenomic layer [13] for C-mVAE

1.7 Study Limitations

- **Snapshot Bias:** ST data represents a static physiological state; cell interactions reflect spatial proximity rather than dynamic causality [6].
- **Visium Resolution (55 μm):** Dependence on SPOTlight deconvolution accuracy in the absence of subcellular CosMx resolution data [15].
- **Sample Imbalance:** Potential sex imbalance in public datasets, explicitly documented following SAGER standards [16].
- **Lack of Wet-Lab Validation:** Experimental validation (qPCR/IHC) remains out of scope for this computational thesis [1].
- **Over-smoothing:** Risk inherent to deep GNNs, mitigated via skip connections [7].

1.8 Feasibility

1.8.1 Data Access

All primary datasets are publicly available with immediate, unrestricted access (GEO, Human Cell Atlas, GTEx) [5, 3, 4]

1.8.2 Identification of the SDGs

SDG 3 s (Good Health and Well-being): By advancing analytical techniques for precision medicine and the early identification of biomarkers in complex diseases.

SDG 5 (Gender Equality): By developing algorithms that correct historical biases in medical data through stratified and conditioned architectures.

SDG 9 (Industry, Innovation and Infrastructure): Providing technological innovation through the use of explainable artificial intelligence (XAI) in bioinformatics infrastructures..

1.9 Plagiarism Prevention

The document will be processed through UOC/UB anti-plagiarism screening (Turnitin). Code will be released under the open-source MIT License on GitHub. References follow APA 7th style.

1.10 Risk Analysis and Mitigation

The following table identifies the principal risks associated with the research, their likelihood and impact, and the corresponding mitigation strategies.

Table 1.1: Risk register and mitigation plan for SexOmics-Spatial TFM

Risk	**Likelihood**	**Impact**	**Mitigation Strategy**
Joint GNN+VAE training divergence	Medium	High	Decouple training: first optimise VAE embeddings; use them as static node features for the GNN in a second stage [10].
Visium resolution insufficient for sex-specific niche detection	Low	High	Switch to CosMx single-transcript data from Campos et al. [5], which provides subcellular resolution without deconvolution.
Sex imbalance in public datasets	Medium	Medium	Apply inverse class weighting during training; report imbalance explicitly following SAGER guidelines [16].
CosMx data not available in GEO at project start	Low	Medium	Fall back exclusively to Visium + SPOTlight deconvolution [15] using GSE210572 as sex-stratified reference.
Over-smoothing in deep GNN layers	Medium	Medium	Introduce skip connections and limit architecture to 3 GAT layers; validated in ST benchmarks [8].
GPU compute cost exceeds budget	Low	Low	Use Kaggle Notebooks (30 h GPU/week, free) as primary alternative to Google Colab Pro; apply NeighborLoader subsampling in PyTorch Geometric.
Physical tissue discontinuity (necrosis, calcification artefacts)	Low	Medium	Replace standard edge-loss with a Gromov–Wasserstein optimal transport cost function for topological alignment across tissue gaps.

1.11 Work schedule

Below is the detailed Gantt chart covering all project milestones.

Table 1.2: Master's Thesis Work Breakdown Structure (WBS) & Effort Estimates

ID	Phase / Deliverable	Task Description	Start Date	End Date	Days	Hours	Dep.
1.1	PEC 1: Work Plan	Define Title, Keywords & Proposal Summary	2026-09-23	2026-09-26	4	12	–
1.2	PEC 1: Work Plan	Context, Justification & Motivation	2026-09-27	2026-09-29	3	10	1.1
1.3	PEC 1: Work Plan	Define General/Specific Objectives, Main Hypothesis & Research Questions	2026-09-30	2026-10-02	3	10	1.2
1.4	PEC 1: Work Plan	Research Strategy, Gender Perspective & SDGs Alignment	2026-10-03	2026-10-05	3	12	1.3
1.5	PEC 1: Work Plan	Work Plan Schedule, Risk Analysis & Final Delivery (PEC 1 Submission)	2026-10-06	2026-10-07	2	8	1.4
2.1	PEC 2: State of the Art	Systematic Literature Search (Scopus, Web of Science, PubMed)	2026-10-08	2026-10-15	8	25	1.5
2.2	PEC 2: State of the Art	Download Spatial Transcriptomics & scRNA-seq Datasets (GEO) + QC	2026-10-16	2026-10-22	7	20	2.1
2.3	PEC 2: State of the Art	Synthesis of Benchmark Methods & Critical Review	2026-10-23	2026-10-29	7	20	2.1

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(continued from Table 1.2)

ID	Phase / Deliverable	Task Description	Start Date	End Date	Days	Hours	Dep.
2.4	PEC 2: State of the Art	Cellular Deconvolution Baseline Execution & Progress Report (PEC 2 Submission)	2026-10-30	2026-11-04	6	15	2.2, 2.3
3.1	PEC 3: Implementation	GNN Architecture Design & Spatial Graph Construction (PyTorch Geometric)	2026-11-05	2026-11-12	8	30	2.4
3.2	PEC 3: Implementation	Biological Sex-Conditioned VAE Integration (FiLM layers)	2026-11-13	2026-11-20	8	30	3.1
3.3	PEC 3: Implementation	XAI Biomarker Extraction & Algorithmic Validation	2026-11-21	2026-11-27	7	25	3.2
3.4	PEC 3: Implementation	Interactive Visualization Dashboard & Implementation Report (PEC 3 Submission)	2026-11-28	2026-12-02	5	20	3.3
4.1	PEC 4: Final Report	Structure Master Thesis Report & Write Introduction/Background	2026-12-03	2026-12-10	8	25	3.4
4.2	PEC 4: Final Report	Compile Experimental Results, Figures & Discussion Synthesis	2026-12-11	2026-12-20	10	35	4.1
4.3	PEC 4: Final Report	Formal Evaluation of Ethics, SDGs & Gender Impact Integration	2026-12-21	2026-12-28	8	20	4.2

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(continued from Table 1.2)

ID	Phase / Deliverable	Task Description	Start Date	End Date	Days	Hours	Dep.
4.4	PEC 4: Final Report	Final Document Formatting (<90 pages), References Audit & PDF Delivery (PEC 4 Submission)	2026-12-29	2027-01-07	10	25	4.3
5.1	PEC 5: Presentation	Virtual Slide Deck Design (Max 20 slides, UOC Official Template)	2027-01-08	2027-01-10	3	12	4.4
5.2	PEC 5: Presentation	Oral Defense Rehearsal & Video Recording (<20 minutes limit)	2027-01-11	2027-01-12	2	10	5.1
5.3	PEC 5: Presentation	Defense Submission (Video File & Presentation Deck Delivery)	2027-01-13	2027-01-13	1	3	5.2
		Total Workload Estimate:			**113**	**367**	

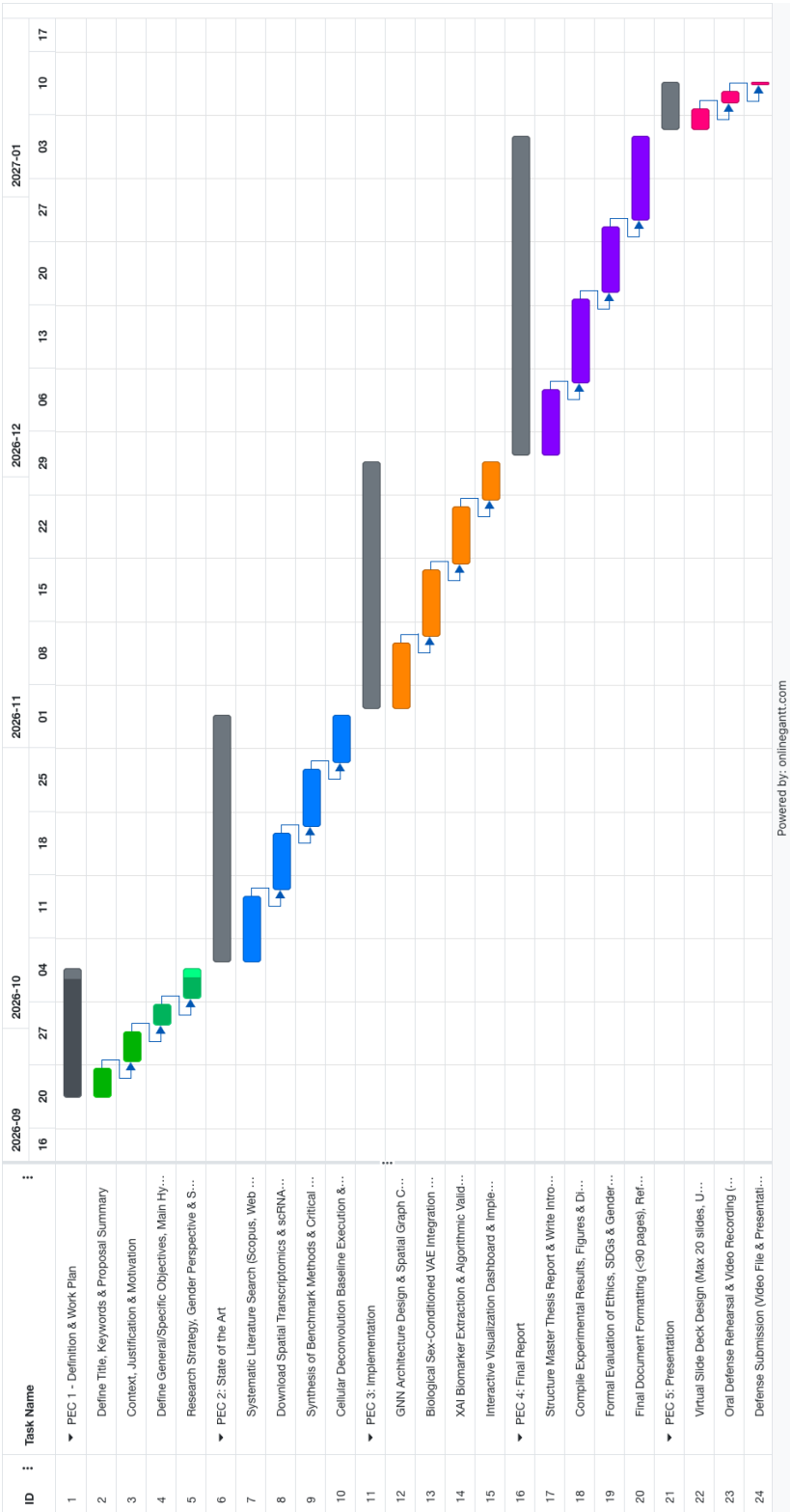


Figure 1.1: Master's Thesis Schedule and Work Breakdown Structure.

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List of sections that are too extensive to include in memory and have a self-contained character (e.g. user manuals, installation manuals, etc.)

Depending on the type of work, there may be no need to add an annex.